

Загрузка и вывод данных

The screenshot shows the Zeppelin Notebook interface with a job titled "DZ". The code defines three DataFrames:

```
%spark.pyspark
from pyspark.sql import functions as F

users = spark.createDataFrame(
    [
        ("u1", "Berlin"),
        ("u2", "Berlin"),
        ("u3", "Munich"),
        ("u4", "Hamburg"),
    ],
    ["user_id", "city"]
)

products = spark.createDataFrame(
    [
        ("p1", "Ring VOLA"),
        ("p2", "Ring POROG"),
        ("p3", "Ring TISHINA"),
    ],
    ["product_id", "product_name"]
)

orders = spark.createDataFrame(
    [
        ("o1", "u1", "p1", 2, 10.0),
        ("o2", "u1", "p2", 1, 30.0),
        ("o3", "u2", "p1", 1, 10.0),
        ("o4", "u2", "p3", 5, 7.0),
        ("o5", "u3", "p2", 3, 30.0),
        ("o6", "u3", "p3", 1, 7.0),
        ("o7", "u4", "p1", 10, 10.0),
    ],
```

The job status is "FINISHED". The bottom of the screen shows a Windows taskbar with a temperature of 2°C and the date 01.03.2026.

The screenshot shows the output of the Spark jobs. The first job displays the contents of the 'users' DataFrame:

```
%spark.pyspark
users.show()

+-----+-----+
|user_id|  city|
+-----+-----+
|   u1   | Berlin|
|   u2   | Berlin|
|   u3   | Munich|
|   u4   | Hamburg|
+-----+-----+
```

The second job displays the contents of the 'products' DataFrame:

```
%spark.pyspark
products.show()

+-----+-----+
|product_id|product_name|
+-----+-----+
|         p1|   Ring VOLA|
|         p2|   Ring POROG|
|         p3| Ring TISHINA|
+-----+-----+
```

Both jobs are marked as "FINISHED". The bottom of the screen shows a Windows taskbar with a temperature of 2°C and the date 01.03.2026.

Zeppelin Notebook Job

Search anonymous

DZ

Took 5 sec. Last updated by anonymous at March 01 2026, 11:39:16 AM.

```
%spark.pyspark
orders.show()
```

SPARK JOB FINISHED

order_id	user_id	product_id	qty	price
o1	u1	p1	2	10.0
o2	u1	p2	1	30.0
o3	u2	p1	1	10.0
o4	u2	p3	5	7.0
o5	u3	p2	3	30.0
o6	u3	p3	1	7.0
o7	u4	p1	10	10.0

Took 1 sec. Last updated by anonymous at March 01 2026, 11:40:52 AM.

```
%spark.pyspark
orders = orders.withColumn("revenue", orders["qty"] * orders["price"])
```

FINISHED

Took 0 sec. Last updated by anonymous at March 01 2026, 11:41:03 AM.

2°C Облачно 11:41 01.03.2026

Задание 1

Took 1 sec. Last updated by anonymous at March 01 2026, 11:40:52 AM.

```
%spark.pyspark
orders = orders.withColumn("revenue", orders["qty"] * orders["price"])
```

FINISHED

Took 0 sec. Last updated by anonymous at March 01 2026, 11:41:03 AM.

```
%spark.pyspark
orders.show()
```

SPARK JOB FINISHED

order_id	user_id	product_id	qty	price	revenue
o1	u1	p1	2	10.0	20.0
o2	u1	p2	1	30.0	30.0
o3	u2	p1	1	10.0	10.0
o4	u2	p3	5	7.0	35.0
o5	u3	p2	3	30.0	90.0
o6	u3	p3	1	7.0	7.0
o7	u4	p1	10	10.0	100.0

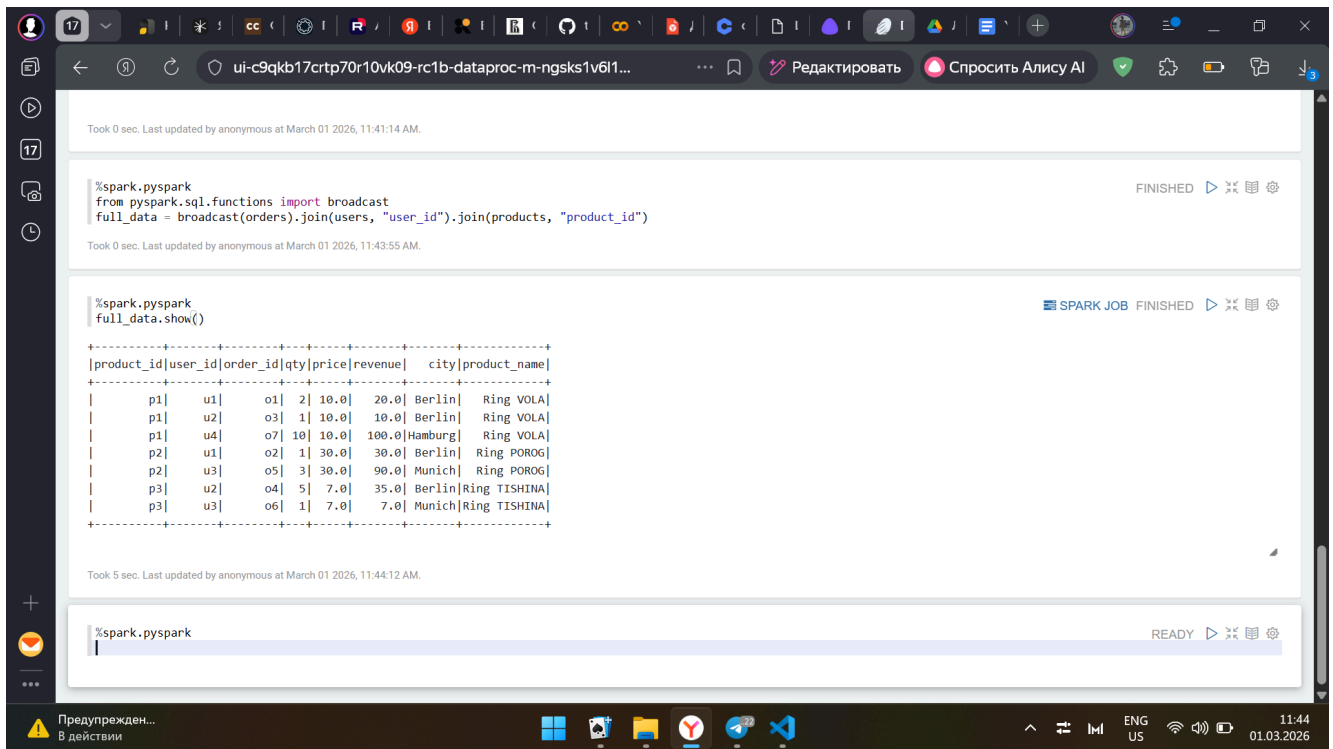
Took 0 sec. Last updated by anonymous at March 01 2026, 11:41:14 AM.

```
%spark.pyspark
```

READY

2°C Облачно 11:42 01.03.2026

Задание 2



Task 0 sec. Last updated by anonymous at March 01 2026, 11:41:14 AM.

```
%spark.pyspark
from pyspark.sql.functions import broadcast
full_data = broadcast(orders).join(users, "user_id").join(products, "product_id")
```

Task 0 sec. Last updated by anonymous at March 01 2026, 11:43:55 AM.

```
%spark.pyspark
full_data.show()
```

SPARK JOB FINISHED

product_id	user_id	order_id	qty	price	revenue	city	product_name
p1	u1	o1	2	10.0	20.0	Berlin	Ring VOLA
p1	u2	o3	1	10.0	10.0	Berlin	Ring VOLA
p1	u4	o7	10	10.0	100.0	Hamburg	Ring VOLA
p2	u1	o2	1	30.0	30.0	Berlin	Ring POROG
p2	u3	o5	3	30.0	90.0	Munich	Ring POROG
p3	u2	o4	5	7.0	35.0	Berlin	Ring TISHINA
p3	u3	o6	1	7.0	7.0	Munich	Ring TISHINA

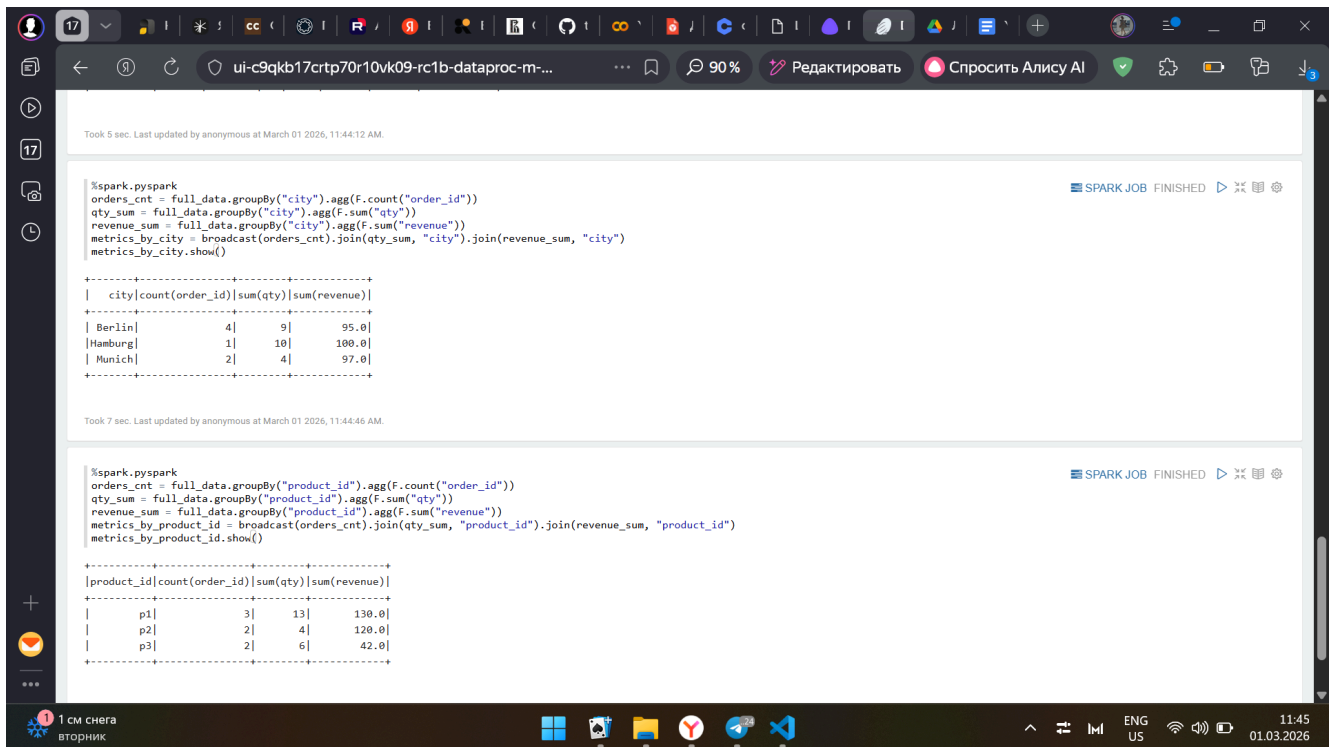
Task 5 sec. Last updated by anonymous at March 01 2026, 11:44:12 AM.

```
%spark.pyspark
```

READY

Предупрежден... В действии

Задание 3



Task 5 sec. Last updated by anonymous at March 01 2026, 11:44:12 AM.

```
%spark.pyspark
orders_cnt = full_data.groupBy("city").agg(F.count("order_id"))
qty_sum = full_data.groupBy("city").agg(F.sum("qty"))
revenue_sum = full_data.groupBy("city").agg(F.sum("revenue"))
metrics_by_city = broadcast(orders_cnt).join(qty_sum, "city").join(revenue_sum, "city")
metrics_by_city.show()
```

SPARK JOB FINISHED

city	count(order_id)	sum(qty)	sum(revenue)
Berlin	4	9	95.0
Hamburg	1	10	100.0
Munich	2	4	97.0

Task 7 sec. Last updated by anonymous at March 01 2026, 11:44:46 AM.

```
%spark.pyspark
orders_cnt = full_data.groupBy("product_id").agg(F.count("order_id"))
qty_sum = full_data.groupBy("product_id").agg(F.sum("qty"))
revenue_sum = full_data.groupBy("product_id").agg(F.sum("revenue"))
metrics_by_product_id = broadcast(orders_cnt).join(qty_sum, "product_id").join(revenue_sum, "product_id")
metrics_by_product_id.show()
```

SPARK JOB FINISHED

product_id	count(order_id)	sum(qty)	sum(revenue)
p1	3	13	130.0
p2	2	4	120.0
p3	2	6	42.0

1 см снега
вторник

Задание 4

The screenshot shows a Databricks workspace with two Spark jobs. The first job, titled "ui-c9qkb17crtp70r10vk09-rc1b-dataproc-m...", has a status of "SPARK JOB FINISHED". It contains the following code:

```
%spark.pyspark
orders_cnt = full_data.groupBy("product_name").agg(f.count("order_id"))
qty_sum = full_data.groupBy("product_name").agg(f.sum("qty"))
revenue_sum = full_data.groupBy("product_name").agg(f.sum("revenue"))
metrics_by_product_name = broadcast(orders_cnt).join(qty_sum, "product_name").join(revenue_sum, "product_name")
metrics_by_product_name.show()
```

The output of the first job is a table with 4 columns: product_name, count(order_id), sum(qty), and sum(revenue). The data is as follows:

product_name	count(order_id)	sum(qty)	sum(revenue)
Ring VOLA	3	13	130.0
Ring TISHINA	2	6	42.0
Ring POROG	2	4	120.0

The second job, also titled "ui-c9qkb17crtp70r10vk09-rc1b-dataproc-m...", has a status of "SPARK JOB FINISHED". It contains the following code:

```
%spark.pyspark
from pyspark.sql.window import Window
from pyspark.sql.functions import row_number, desc, sum, col

top_revenue = full_data.groupBy("city", "product_name").agg(sum("revenue"))
w = Window.partitionBy("city").orderBy(desc("sum(revenue)"))
top_revenue = top_revenue.withColumn("rn", row_number().over(w)).filter("rn <= 2")
top_revenue.show()
```

The output of the second job is a table with 4 columns: city, product_name, sum(revenue), and rn. The data is as follows:

city	product_name	sum(revenue)	rn
Berlin	Ring TISHINA	35.0	1
Berlin	Ring POROG	30.0	2
Hamburg	Ring VOLA	100.0	1
Munich	Ring POROG	90.0	1
Munich	Ring TISHINA	7.0	2

Задание 5 и 6

The screenshot shows a Databricks workspace with two Spark jobs. The first job, titled "ui-c9qkb17crtp70r10vk09-rc1b-dataproc-m-ngsks1v6l12a59ci-8890...", has a status of "SPARK JOB FINISHED". It contains the following code:

```
%spark.pyspark
path = "s3a://for-study/mart_city_top_products/"
top_revenue.write.mode("overwrite").parquet(path)
spark.read.parquet(path).show()
```

The output of the first job is a table with 4 columns: city, product_name, sum(revenue), and rn. The data is as follows:

city	product_name	sum(revenue)	rn
Berlin	Ring TISHINA	35.0	1
Berlin	Ring POROG	30.0	2
Hamburg	Ring VOLA	100.0	1
Munich	Ring POROG	90.0	1
Munich	Ring TISHINA	7.0	2

The second job, also titled "ui-c9qkb17crtp70r10vk09-rc1b-dataproc-m-ngsks1v6l12a59ci-8890...", has a status of "SPARK JOB FINISHED". It contains the following code:

```
%spark.pyspark
path = "/tmp/sandbox_zeppelin/mart_city_top_products/"
top_revenue.write.mode("overwrite").parquet(path)
spark.read.parquet(path).show()
```

The output of the second job is a table with 4 columns: city, product_name, sum(revenue), and rn. The data is as follows:

city	product_name	sum(revenue)	rn
Berlin	Ring TISHINA	35.0	1
Berlin	Ring POROG	30.0	2
Hamburg	Ring VOLA	100.0	1
Munich	Ring POROG	90.0	1
Munich	Ring TISHINA	7.0	2

